UNIX

1) what is unix & explain Features of unix os ?

Ans.

• unix :

- "unix is one of The oldest operating systems, which is a multi-user and multi-tasking system."

- The unix os is written in c programming language.
- "under unix, The operating system consits of many utilities along with The master control program, called as The kernel."
- in mid 1960 MIT, Bell Labs and other organization made a multics but it is complex so it is donot work any other, after some reachers made multics To unics who is Dennis Ritchie, ken Thompson, Brain ~~Kerni~~ kernighan after it was renamed as unix.
- in 1972, The unix was re-written in c language, migrating from assembly level language.

→ Features of unix os :

1) multi user capability
2) multi tasking capability
3) Inter process communication
4) Security
5) Portability
6) open system
7) windowing system
8) system calls and libraries
9) Programming facility

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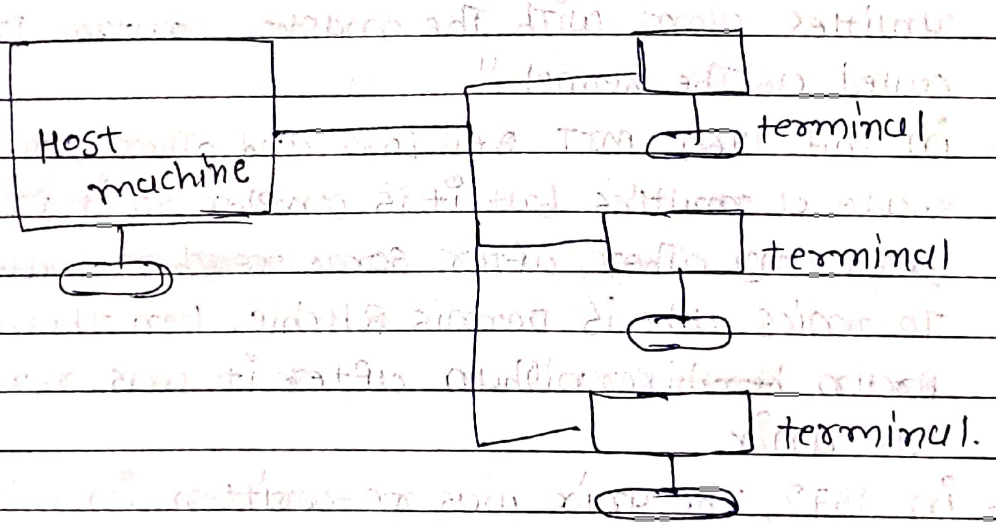
10) networking

1) multi-user capability :

- multi-user features allows user to use The resources of main computer.
- The Resources of host machine are available to all users.



a) dumb Terminal
- The terminal consits of only a keyboard and a monitor is known as dumb Terminal.

b) Terminal Emulation :
- This terminal consits of keyboard and monitor with its own microprocessor, memory, and hard disk drives.

c) Dial-in Terminal :
- If The host machine is at a remote location Then Dial-in Terminal is used.

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2) multitasking capability :

- Another main features of unix os is multitasking.
- multitasking is an important component of the Process management system.

3) Inter process communication :

- This features allow user to communicate with users.
- allow to user pass data, exchange email etc.

4) Security :

- unix provide Three types of security : i) system level
ii) directory level
iii) file level

5) portability :

- unix system is written in a high level language.
- The code can be changed and compiled on a new machine.

6) open system :

- This features is allows to open system to stored file in a seperate drive device.
- A seperate device can also be added by creating a file for it.

7) windowing system :

- unix has a pretty weak user interface in that it was command-driven.

8) system calls and libraries :

- ~~unix is highly programmable; it was~~ unix is written in c language.
- There are many commands for handle special functions that is called system calls.

9) programming facility :

- unix is highly programmable; it was designed for programmer, not a casual end user.

10) networking :

- unix was not originally a networking system.
- This concept was added to unix system after a split between BSD unix and AT & T unix.

2) what is unix & explain system ~~archi~~ structure of unix ?

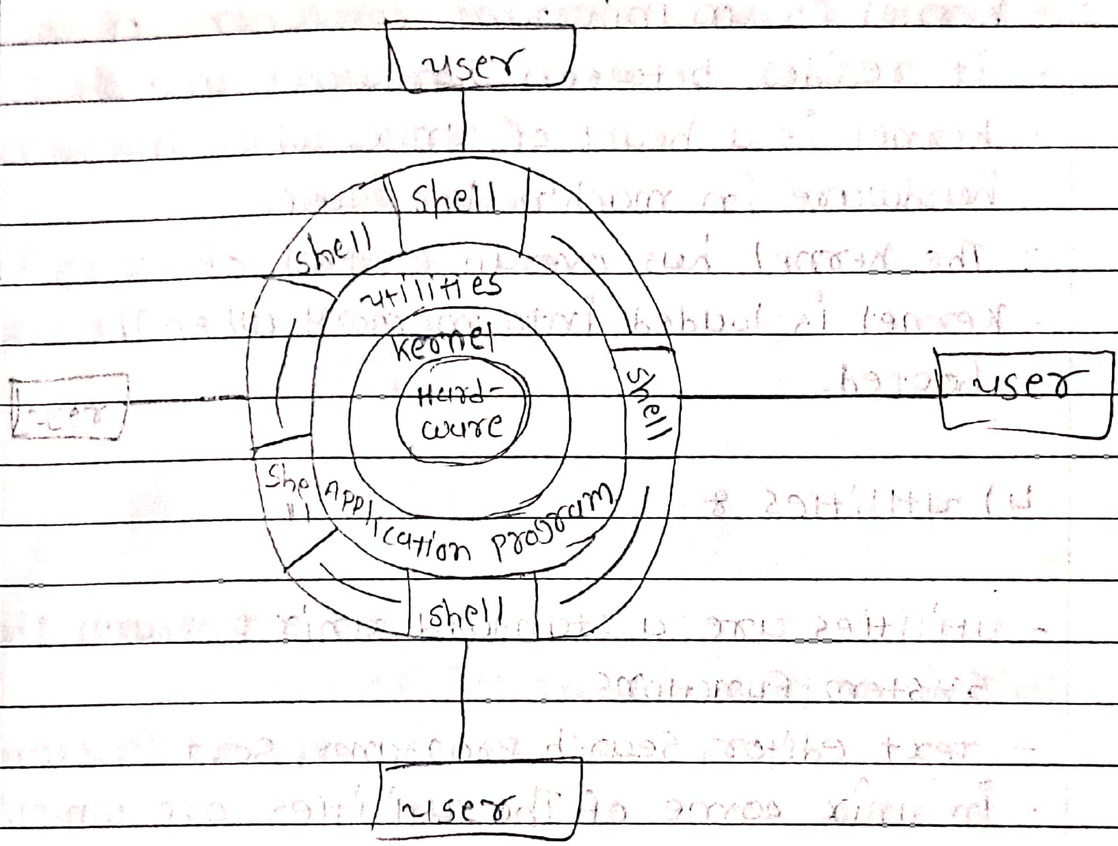
Ans.

• unix :

- "unix is one of the oldest os, which is a multi-user and multi-tasking system."

→ unix system organization :

- unix is a layered operating system.
- There are Three major components of unix operating system.
- The function of unix is managed in Three levels.



1) Hardware :

- The innermost layer is The hardware That provides The services for The os.

2) shell :

- The shell is one outer layer.
- it is a command interpreter which interprets The command supplied by user at shell prompt.

- it is actually an interface between user and The kernel.
- The shell is represented by sh, csh, ksh, bush.

3) kernel :

- kernel is an important component of unix os.
- it resides between hardware and shell.
- kernel is a heart of unix, which interacts actual hardware in machine language.
- The kernel has overall control of everything.
- kernel is loaded into memory when The system is booted.

4) utilities :

- utilities are a standard unix programs That perform system functions.
- Text editor, Search programs, Sort Programs etc.
- in unix some of The utilities are complicated Applications.

5) Applications :

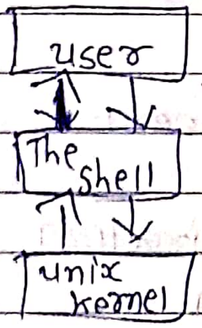
- The programs written by system Administrator, professional programmers, or users are known as applications.
- Applications are not a standard part of unix system.
- They provide an extended capability to The system

3) what is shell & explain its features & its type ?

Ans.

• " shell " :-

- " shell is most widely used utility program on all unix systems."
- it is louded in memory when a user logs in.



→ Types of shell :-

1) Bourne shell :-
- This is one of The most widely used shells in The unix world.

2) C shell :-
- Bill Joy developed c shell at The university of california.
- Its commands were supposed to like c statement.

3) Korn shell :-
- it is very powerfull shell.
- it is a superset of Bourne shell.

4) bash shell :-
- it is an enhanced version of The Bourne shell.
- it is used in Linux environment.

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5) tcsh shell :-
- tcsh is pronounced as tee-cee shell.
- it is used in linux environment.

6) Pdksh shell :-
- Pdksh stand for public domain korn shell.

→ Features of shell :-

1) Interactive environment.
2) shell scripts
3) Input / output redirection.
4) piping mechanism
5) metacharacter facilities
6) Background Processing
7) Customized environment.
8) programming Language constructs.
9) shell variables.

4) what is kernel & explain kernel Architecture of unix os ?

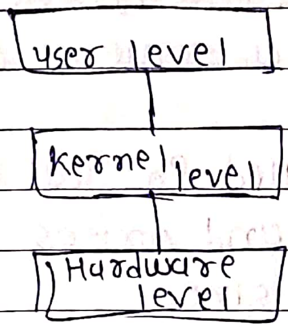
Ans.

• kernel :-

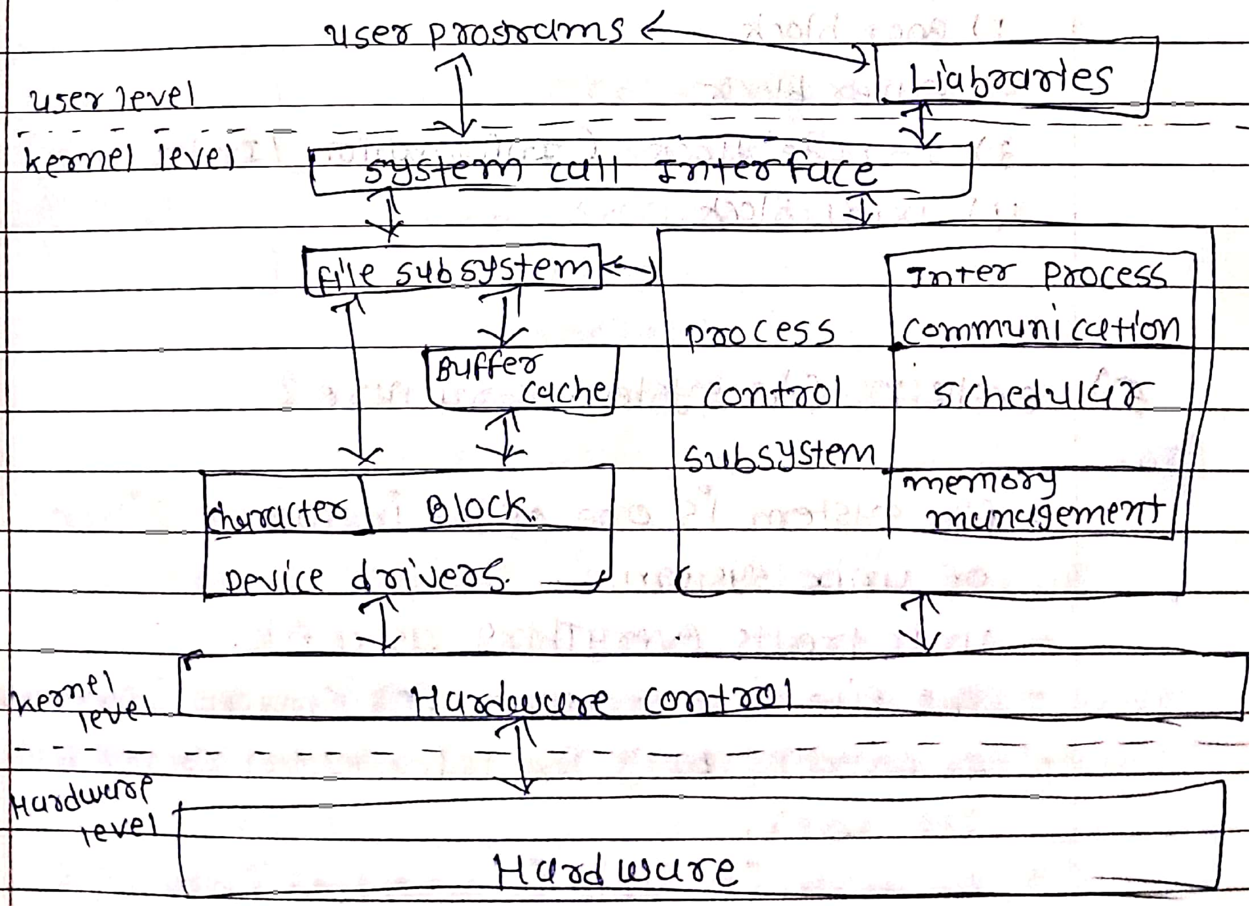
- " kernel is a heart of unix, which interacts with the actual hardware in machine language "

→ kernel Architecture in unix os :-

- The unix os has Three levels : user level, kernel level and hardware level.



→ diagram of kernel unix os architecture :



5) explain file system components ?

Ans.

- when you create a file system on a disk, it is organized into a sequence of logical blocks.
- This blocks is called Physical block.
- The kerel reads and writes data using 4 different block size.

Boot Block	Super Block	I-node block	Data Block

1) Boot block.
2) Super block
3) I-node block (Information / Index node)
4) Data block.

1) boot block :

- boot block is The smallest part of The unix file system.
- boot block is normally reserved for booting procedures.
- The control hands over to The bootstrapping program.

2) The super block :

- The super block or block 1 is The balance sheet of every unix file system.

- it mainly contains:
  - The size of file system
  - last time updating
  - size of logical block of the file system

3) I-node (Information / Index) Block:

- we know ~~all~~ that all entities in unix are treated as files.
- it is store relevent details. These details are:
  - file type
  - number of links
  - numeric uid of owner
  - numeric Guid of owner
  - File access permission
  - Date and time of last modification, changes, and access.

- I-node is accessed by a number called I-node number.

4) Data block.:

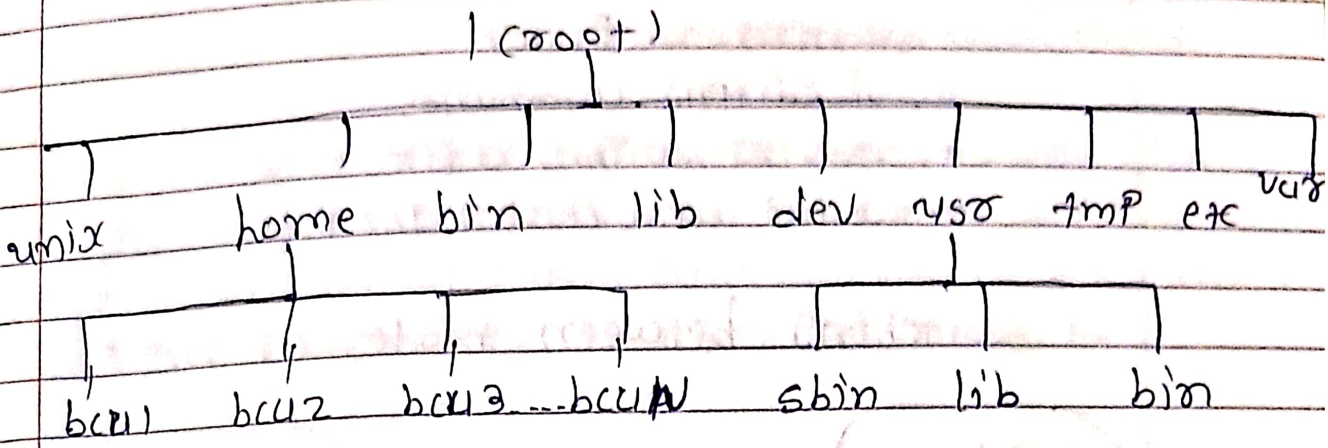
- The remaining space of the file system is taken up by data blocks or storage blocks.
- Data blocks is the largest part among other parts.
- These blocks store the contents of the file in the case of a regular file.

6) explain file system structure?

Ans.

- file system is one of The important pillar of unix system.
- unix treats everything as a file.
- The file is a container for storing information
- It contains only The information stored by The user.
- All data in unix is organized into files.
- All files are organized into directories.

→ diagram of file system :-



→ classification of unix file system :-

- ordinary files
- Directories
- special files
- pipes
- sockets
- symbolic links

7) explain vi editor in breifly ?

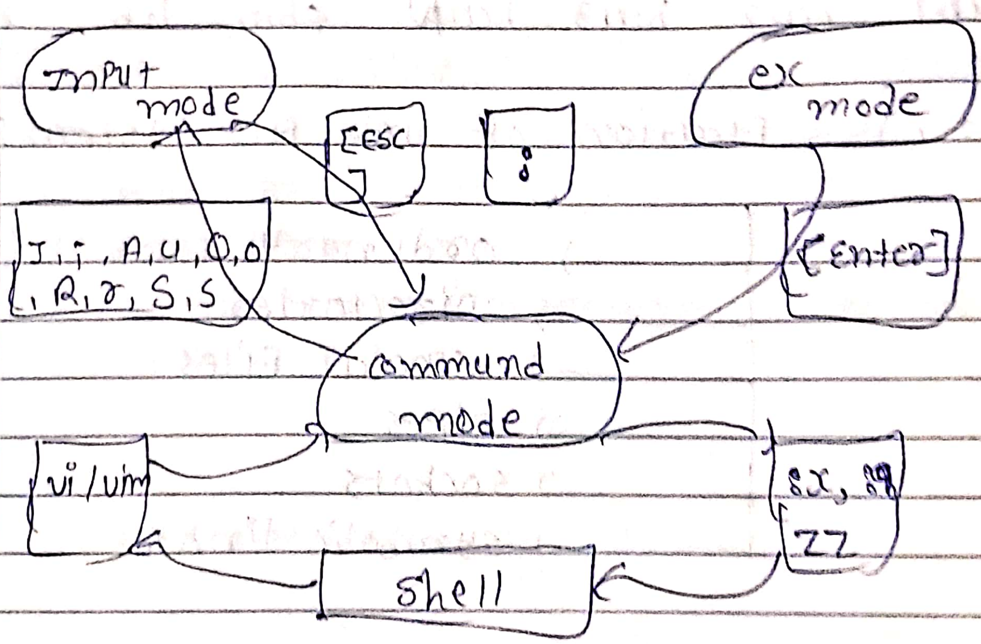
Ans.

- vi editor is The most popular and classic text editor in The linux famiry. Below, are some reasons :-
  - i) It is available in almost all linux distributions
  - ii) it works same platforms and distributions
  - iii) it is user-friendly.

- To work on VI editor, you need to understand its operation modes.

- command mode
- insert mode
- starting vi Editor
- vi editing commands
- moving within a file
- saving and closing the file

→ switching between modes of vim :



~~command mode~~ : vi editing commands :

i – insert at cursor
a – write after cursor
A – write at the end of line
ESC – Terminate insert mode
u – undo last change
U – undo all changes
o – open a new line
dd – delete line
r – Replace character

x - Delete the character at the cursor.

8) explain shell's interpretation at prompt?

Ans.

- when user enters command on command-line, shell perform some internal ~~spac~~ steps before executing the command and directs it execution by shell a utility or other programs.

1) Parsing :-
- The shell divides the command line into words, if it is not quoted or escaped.

2) variable evaluation :-
- The shell is looking for variable name (s).

3) command substitution :-
- command executed of back quotes by the shell and output is substituted as argument.

4) Redirection :-
- The shell scans for the characters >, < and >> and open a file associated with this character in a particular mode.

5) wild - card interpretation :-
- command line for wild - cards or filename expansion characters such as *, ?, [and].

6) PATH evaluation :-
- Search a command into the sequence of directories stored in it.

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9) what is booting sequence and init process ?

Ans.

→ When the system starts, it perform following steps :-

1) Boot loader is loaded :-

- The Rom BIOS reads the boot loader from boot block and loads it into a RAM.

2) Kernel is loaded :-

- The kernel program is called by the user.

3) Kernel initialization takes place :-

- It involves memory tests, device driver, swapper scheduler process and init process.

4) The init program :-

- This process has the PID which is the second process of the system.

5) The getty process :-

- The getty process is invoke program which is establishes a communication with unix termin

6) The login program :-

- once the user types login name and passwor, getty transfer control to a login program.

→ how to the sequence processes leads to shell proce

Init → getty → login → shell.

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-) how To The sequence processes leads to shell Process

Init -) getty -) login -) shell.

10) explain berifiy file access permission ?

Ans.

- In unix, all files and directories have permission.
- There Three file level and directory level permission: read, write and execute.
- symbolically Represented :

• Read Permission - r
• write Permission - w
• execute permission - x

- They have the following octal values :

• Read Permission - 4
• write permission - 2
• execute permission - 1

Permission	symbolic code	octal value
No Permission	-	0
Execute	x	1
write	w	2
write and execute (2+1)	wx	3
read	r	4
Read and execute (4+1)	rx	5
Read and write (4+2)	rw	6
Read, write and execute (4+2+1)	rwx	7

-) There are two type of permission : 1) Relative Permission
2) Absolute Permission

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1) Relative Permissions :
- when changing Permissions in a Relative manner, chmod changes only the Permissions specified in the command line.
- syntax :

chmod [category operation permission] Filenames

→ example :
chmod u+x file1.txt.

- remove the execute Permission from all and then to assign read Permission to group and others.

2) Absolute Permissions :
- Irrespective of existing Permissions for a file, of we may need to assign a new set of permission.
- we wish to set all nine Permission bits explicitly.
- as per Permission given digit as below -
  - Read Permission - 4
  - write permission - 2
  - execute Permission - 1

10) explain Arithmetic expression evaluation?

Ans.

- we will discuss some of the unix commads That are used to evaluate ~~exit~~ arithmetic expression.

1) expr :

- It performs arithmetic operations on integers and string manipulation on strings.
→ syntax : expr [expression]

expression	meaning
ARG1 + ARG2	sum of arg 1 and arg 2
ARG1 - ARG2	difference of arg1 and arg2
ARG1 * ARG2	Product of arg1 and arg2
ARG1 / ARG2	quotient of Arg1 divided by Arg2
ARG1 % ARG2	Reminder of arg1 divided by arg2

2) bc (basic calculator) :

- bc command is used for command line calculator
- The bc command and expr command for doing arithmetic calculations.

→ example :

Input : $ echo "10+5" | bc
Output : 15

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3) using $ ((....)) :
- in bash $() is used for command substitution.
- (..) is used for writing Arithmetic expression.
-> syntax : $ ((expression))
e.g. z = $ (($x + $y))

4) factor :
- This is another math-Related command available in unix.
-> syntax : factor [number(s)]....

11) explain shell metacharacter ?

Ans.

- There are different types of metacharacters supported by unix shell.
- we can characterize metacharacter as follows

1) filename substitution (pattern matching - The wild cards)
2) Redirection
3) piping
4) Quoting and escaping.

1) filename substitution :

- The metacharacter That are used to construct the generalized pattern for matching filenames belong to a category called wild-cards.
i) Asterisk (*)
ii) Question-mark (?)
iii) Character class []

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2) Redirection :

- in ~~ux~~ unix, There are Three pre-assigned standard stream such as standard input, standard output and standard error.
- Redirection allows user to change the default behavior of standard stream.

3) piping :

- unix has a feature by which filters and other commands can be combined.
- it is denoted by vertical bar.
→ syntax :

command 1 | command 2 | ... | command N

- There is no restriction on number of commands used in pipeline.

4) Quoting and escaping :

- quotes and escape are used to change The predefined meanings of meta-characters.
- ~~There~~ They are Three ~~ty~~ types :

i. Backslash
ii. a pair of double
iii. a pair of single quotes.

12) explain control structure in unix OS :

Ans.

- control structure Alters The flow of execution of shell script.
- There are two control statement :
  1) If statement
  2) case - esac statement.
  3) loop control

1) If statement :

- It takes decisions, depending on of certain condition.
- There are Three types of IF statement :
  i. IF - Then - fi statement
  ii. IF - Then - else - fi statement
  iii. IF - Then - elif - else - fi statement.

(i) IF - Then - fi statement :

→ syntax :

```
IF control command
Then
        statement (s)
fi
```

(ii) if - Then - else - fi statement :

→ syntax :

```
IF control command
Then
        statement (s)
else
        statement (s)
fi
```

(iii) if - Then - elif - else - fi statement :-
→ syntax :-
If control statement/command 1
Then
statement(s)
elif control command 2
Then
statement(s)
else
statement (s)
fi

2) The case statement :-
- The case statement is used for multi level Statement.
→ syntax :-

```
case expression / variable in
    Lable1 1
        Statement (s)
        ;;
    Label 2
        statement(s)
        ;;
    Label N
        statement(s)
        ;;
    [*)
        Statement(s)
        [;;]]
esac
```

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3) Loop control structure :-
- A loop involves repeating some portion of the program either a specified number of times or until a particular condition is being specified.
- They are :-
    i. while loop
    ii. until loop
    iii. for loop.

i) while loop :-
→ syntax :-
```
while control-command
do
        statement (s).
done
```

ii) until loop :-
→ syntax :-
```
until control-command
do
        statement (s)
done
```

iii) for loop :-
→ syntax :-
```
for control-variable in v1 v2 ..... vN
do
        statement (s)
done
```

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13) what is process status & explain it ?

Ans.

- The command reads through the kernel's data structures and process tables to fetch the characteristics of processes.

→ PS options :

A) -f (full listing)
B) -u (displaying process of a user).
C) -a (displaying all user processes)
D) -e or -A (system processes)

14) what do you mean of grep utility ?

Ans.

- The grep stands for globally search a regular expression and point it, it is also known as pattern matching utility.
- The pattern that is searched in the file is referred to as the regular expression.

→ Syntax :

grep [options] pattern [filename(s)]

- it is use to select and extract lines from a file and point value only those lines that match a given pattern.
- in the above syntax square bracket indicates optional part.

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- without a filename grep expects standard input.

→ example of simple regular expression:

RE	meaning
A	All lines contain character "A".
unix	All lines contain pattern "unix".

~~→ exam~~ e.g.:

$ cat demo
Hello
Good morning
How are you all?
Testing tab

→ metacharacter of grep command:

character	use
*	zero or more occurrences of previous character.
ch*	zero or more occurrences of character ch.
. (dot)	except new-line character.
.*	nothing or any number of characters
bash$	bash at the end of line
^bash$	bash as the only word in line

→ Options of grep command :

i) -c (count) :
- it is count line in file.

ii) -l (list) :
- it is display only names of files.

iii) -n (number) :
- It can be used for line numbers.

iv) -v (inverse) :
- select all lines but not contains line pattern.

v) -i (ignore) :
- it ignores case in pattern matching.

vi) -h (hide) :
- it omits filenames when handling multiple files.

• grep family :

- There is a small family of grep utility which includes egrep and fgrep.
- There is two member of grep family :

1) egrep
2) fgrep

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1) egrep :
- egrep is extends from grep.
- it was invented by Alfred Aho.

2) fgrep :
- fgrep stands for fixed / fast grep.
- This is unlike in egrep, which uses The "|" to delimit two expressions.

→ limitation of grep family :-

- it can't used to add, delete or change a line.
- it can't used to point only part of a line.
- it can't read only part of a line.

15) what is sed utility ?

Ans-

- The term sed stands for stream editor and was designed by Lee mcmohan.
- stream editor i.e., sed is derived from ed, known as line editor.

→ syntax :

sed [options] instruction [filename(s)]

- Addresses :

✓ line address can be a single line, or a set of line or range of life.

Address	meaning
5 command	it applies command on 5^{th} line
\$ command	it applies command on last lin
15 command	it applies command on 15^{th} line

→ sed commands :

- The sed support several commands.
- commands are used to apply action on specified lines
  i) print
  ii) quit
  iii) line number
  iv) modify
  v) files
  vi) substitute.

16) explain awk utility :

Ans.

- awk name comes from the initial letter of Three author : Alfred Aho, Peter weinberger, and Brian kernighan.

- The awk is a powerfull programming language designed as utility.
- it also accepts regular expression for pattern matching.

- it has an ability to generate formatted Reports
- it allows The user to specify instructions in a scripts.

→ syntax :-

awk option 'instruction' filename(s)

- only selection criteria or an action or a Combination of both is available.
- selection criteria are not specified then for each record.

→ output statements :-

A) Print statement
B) Printf statement

→ regular expressions :-

re	meaning
.*	zero or more occurrences of any char.
a*	" " " " " " 'a'.
.	matches any single character
a+	one or more occurrences of char 'a'.
a?	zero or one occurrences of char 'a'.

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17) explain write, wall, mesg command ?

Ans.

• write command :

- write command allows user to communicate with other user.
- it is on-line communication command.
- It sends a message to another user who is currently connected to the unix system.

-) syntax :

write user [ttyname]

-) example :

$ write bca 10 [enter]

-) There are two requirements for smooth write operations :
1) receiver must be logged into the system
2) The receiver Terminal is writable.

• wall command :

- It sends a message on the terminal of all the users who are currently logged in to the unix system.
- This command is specially designed for the super user.

-) syntax :

wall [message]

→ example :

$ wall "Good morning Students"

• mesg :

- it controls write access to your Terminal by others.
- it is allow or disallow other users to write to your terminal.

→ syntax :

mesg [y/n]

→ example :

$ mesg
is y

option	meaning
Y is y	it allow write access to your Terminal
N is n	it ~~allow~~ disallow write access your Terminal.

18) explain mail, motd, news command :

Ans.

• mail command :

- it is off-line communication command.
- it sends and receives mail.
- sends a mail even if They are not logged in unix system.

→ system :-

mail [-s subject] [-c cc-addr] [-b bcc-addr]
to-addr1 to-addr2.....

→ mail command :-

i) -s (subject)
ii) -c (carbon copy)
iii) -b (blind carbon copy)
iv) -f (reads content of mbox)
v) -u (it is equivalent of -f)

• motd command :-

- it is a special file resides in /etc directory.
- it is a used for send offline message.
- The content message is display any user logged into the system.
- The motd stands for the "message of the day"
- it is only used for super user.
- it gets executed every time the user log in.

• news command :-

- This command is invoke by user to read any message that is sent by the system Administrator.

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→ syntelx :

news : mess1 mess2 mess3 _ _

→ exemple :

$news

no news

→ news commands :

i) -n (name)
ii) -s (still not been read)
iii) -a (all)

* 2 marks Questions

1) what is variable & its type ?

Ans.

• Shell variable :-

- "A variable created within shell is known as shell variable."
- A variable is nothing more than a pointer to the actual data.

→ There are two types of variables :-
   1) user defined variable
   2) system [environmental variable

2) what is argument processing ?

Ans.

- we understood how to be carefull when we use certain non alphanumeric characters in variable names.
- These variables are reserved for specific functions

~~Nest~~ variable.	meaning
$0	filename of current script
$n	positional parameters.
$#	number of arg supplied to a script

$*	all arg are double quoted
$?	last command executed
$$	Process of The current shell
$!	process of last background command

3) types of mechanism of process creation ?

Ans.

There are Three types of mechanism of process creation :- 1) fork
2) exec
3) wait

4) ~~explain~~ difference b/w wait or not wait ?

Ans.

No./Topic	wait	not wait
lock of monitor	release The lock of monitor	don't Release lock of monitor
present	Present in object class	Present in Thread class
method	non static method	it is static method.
Time unit	Does not accept specific time unit	Accepts time duration in miliseconds

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5) explain head and tail command ?

Ans.

• head :

- head command is used for display few lines of a file.
- head command is used when you watch preview of a file without open entire file.

• tail :

- tail command is used for display few lines of a file.
- tail command is used for when you watch last preview of a file without open entire file.

6) give the list of online and offline command ?

Ans.

• online command :

1) write
2) wall
3) mesg

• offline command :

1) mail
2) motd
3) news

3) give The list of communication command?

Ans. There are so many communication command :-

1) write
2) wall
3) mesg
4) mail
5) motd
6) news